

Convolutional Neural Networks (CNN)

INTELLIGENT SYSTEMS

ESCUELA POLITÉCNICA DE INGENIERÍA DE GIJÓN



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What is a Convolutional Neural Network (CNN)?

Basic Concepts of CNNs

CNN Model Architecture

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References

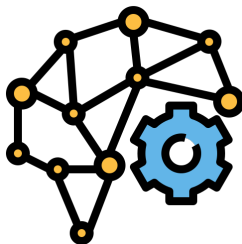
What is a Convolutional Neural Network (CNN)?

What is a Convolutional Neural Network (CNN)?

Convolutional Neural Networks, or **CNNs**, are a class of **deep neural networks** specialized in processing grid-like data (e.g., images), but also text and audio.

Inspired by human vision, **CNNs** excel at:

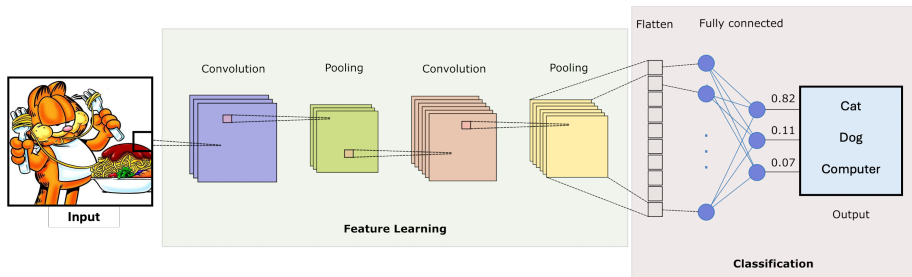
- **Automatic feature extraction**
- **Reducing parameters**
- **Pattern recognition.**



What is a Convolutional Neural Network (CNN)?

CNNs have revolutionized *AI*, achieving state-of-the-art results in tasks like:

- **Image Recognition:** Facial recognition, medical imaging, self-driving cars
- **NLP:** Text classification, sentiment analysis
- **Generative AI:** Deepfakes, artistic style transfer.



How does a CNN recognize a cat? Let's find out!

Basic Concepts of CNNs

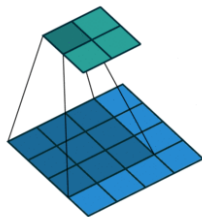
Understanding Filters & Kernels

What is a **filter**?

- A small matrix (**kernel**) used to detect patterns in an image
- Example: Edge detection, sharpening, blurring.

How does it work?

- The **filter** slides over the image and performs element-wise multiplication
- Produces a feature map highlighting detected patterns.



3x3 filter

Convolution Operation

Mathematical formulation for an input x and kernel w :

$$y(i, j) = (x * K)(i, j) = \sum_{m=0}^{m-1} \sum_{n=0}^{n-1} x(i + m, j + n) \cdot w(m, n)$$

where:

- $x(i, j)$ represents the input image pixel value at position (i, j)
- $w(m, n)$ is the filter (**kernel**) value (**weight**) at position (m, n)
- $y(i, j)$ is the output feature map at position (i, j) .

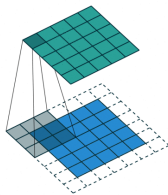
Padding & Stride in CNNs

The output shape of the convolutional layer is determined by the shape of the input and the shape of the convolution kernel.

Two important hyper-parameters that affect to the dimensions of y :

Padding

- Adds extra pixels around the input to preserve spatial dimensions
- Ensures that when the kernel is applied, the output feature map remains the same size as the original input.

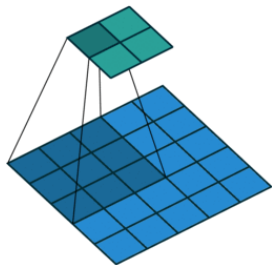


3x3 filter with padding of 1

Padding & Stride in CNN

Stride

- Controls how much the kernel shifts over the input image
- Output feature map will have the same spatial dimensions as the input.



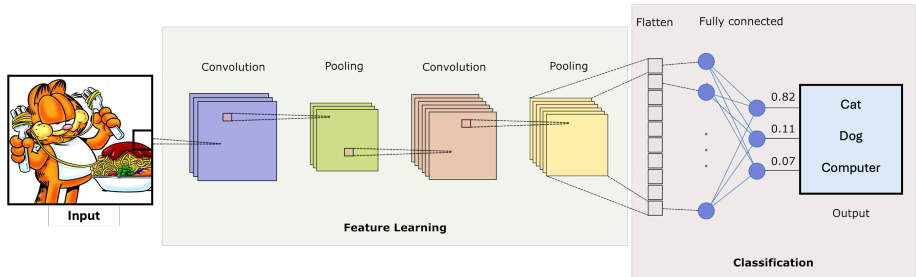
3x3 filter with stride of 2

CNN Model Architecture

CNN Model Architecture

A Convolutional Neural Network (CNN) is comprised of four main layers:

- i) **Input Layer**
- ii) **Convolutional Layer**
- iii) **Pooling Layer**
- iv) **Fully Connected Layer.**



CNN Model Architecture

Input Layer

The **Input Layer** is where the data is fed into the network:

- Receives the raw data (e.g., image), and processes it before passing it to the subsequent layers
- Generally represented as 3D matrices: **height x width x channels** (e.g., 32x32x3 for a color image with 3 channels: red green, and blue)
- The input data may undergo normalization to improve the network's training efficiency.

Convolutional Layer

The **Convolutional Layer** is responsible for detecting features from the input data, such as edges, textures, and patterns:

- **Kernels:** Slide over the input image to detect specific features by computing dot products at each position
- **Feature Maps:** Generate feature maps that highlight the detected features, such as edges or corners
- **Padding + Stride:** Maintain spatial dimensions + define filters step size
- **Activation Function:** An activation function (typically **ReLU**) is used to introduce non-linearity into the network.
ReLU keeps positive values as they are and replaces negative values with zero, helping the network learn complex patterns more efficiently.

The **Pooling Layer** reduces the spatial dimensions of the feature maps, making the network more efficient and helping to prevent overfitting:

- **Max Pooling:** The most common operation, where the maximum value from a defined window (e.g., 2×2) is selected, preserving important features
- **Average Pooling:** Instead of the maximum, it takes the average value from the window, though it's used less frequently.

Fully Connected Layer

The **Fully Connected Layer** is where the network makes final predictions based on the features extracted by the previous layers:

- Each neuron in the **fully connected layer** is connected to every neuron in the previous layer, forming a **dense network**
- The output from the previous layers is **flattened** (i.e., converted into a 1D vector) before being passed into the FC layer
- The FC layer combines the extracted features and performs the final decision-making, **output probabilities**.
Typically using a **sigmoid** (binary) or **softmax** (multi-class) activation function for classification tasks.

CNN Model Training

CNN Model Training

Training a **CNN Model** involves optimizing its weights (parameters) to minimize errors and improve prediction.

Key steps of the **training**:

- i) **Forward Propagation**: Input passes through the model → **Predictions**
- ii) **Loss Calculation**: Compares predictions with actual labels
- iii) **Backpropagation & Optimization**: Computes gradients using the loss function and update weights via **Gradient Descent**
- iv) **Iterations & Convergence**: Repeats until loss is minimized and model performance stabilizes
- v) **Validation & Regularization**: Evaluates on a validation set to prevent overfitting (Dropout, Batch Normalization, Weight Decay)
- vi) **Evaluation**: Performance metrics (accuracy, MSE, etc.).

Parameters in CNN Model Training

Different **parameters** and **hyperparameters** are used at each **training** stage:

i) **Forward Propagation:**

Layer Hyperparameters: Filters (size/number), padding, stride (CONV/POOL), units (FC)

ii) **Loss Calculation:**

Loss Function: Cross-Entropy (classification), MSE (regression)

iii) **Backpropagation & Optimization:**

Trainable Parameters: Weights (CONV, FC layers)

Optimizer: SGD, Adam, etc.

Optimization Hyperparameters: Learning rate, momentum

iv) **Iterations & Convergence:**

Training Hyperparameters: Batch size, number of epochs.

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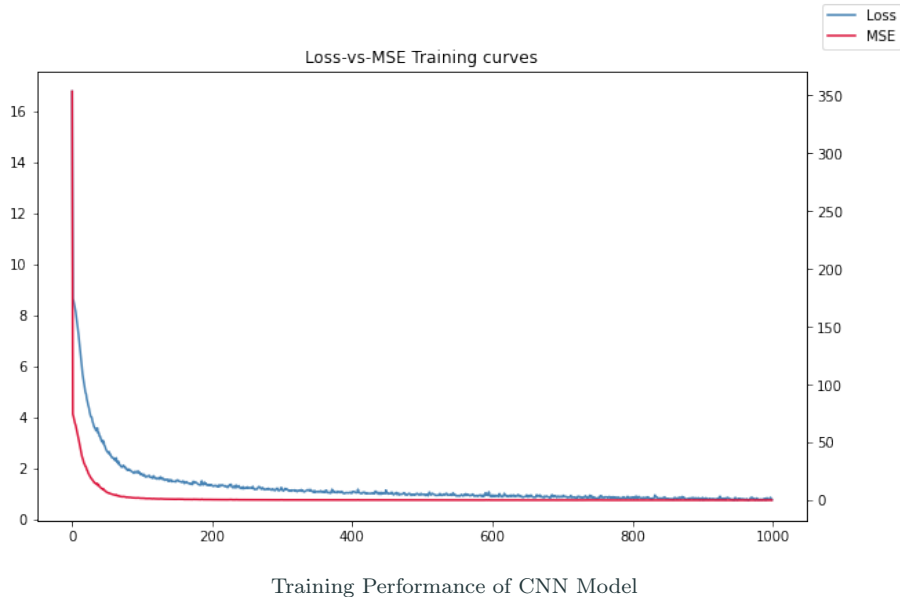
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CNN Training: Loss & Performance Curves over Epochs



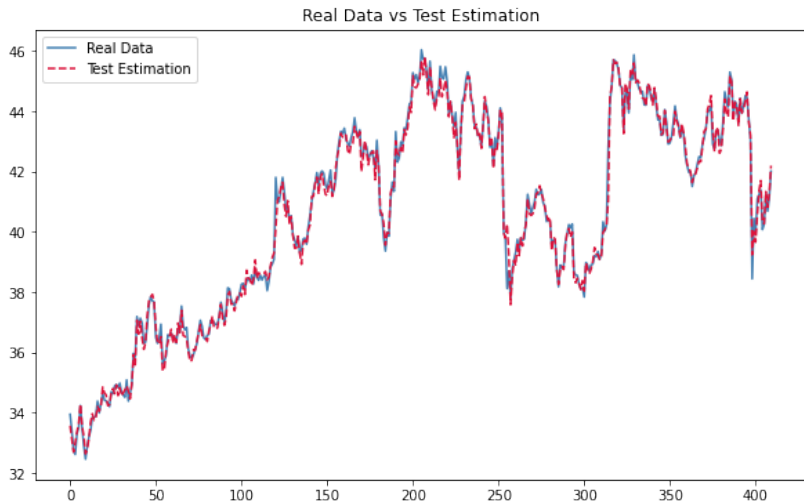
Applications of CNNs

Applications of CNNs

Convolutional Neural Networks (**CNNs**) have a wide range of applications, a few ones:

- **Image Classification:** Medical imaging, autonomous vehicles
- **Object Detection:** Facial recognition, security systems
- **Segmentation:** Medical image analysis, satellite imaging
- **NLP:** Sentiment analysis, voice assistants
- **Time Series Forecasting:** Stock prices, weather forecasting
- **Video Analysis:** Surveillance, autonomous driving
- **Generative Models:** Style transfer, image super-resolution, video games
- **Anomaly Detection:** Machinery failure detection, credit card fraud.

Time Series Forecasting - Stock prices



CNN Model - Stock Market Price Estimation

Generative Models - Style Transfer



(a) content image



(b) style image



(c) result image

CNN Model - Style Transfer

NLP - Sentiment analysis



CNN Model - Sentiment Analysis

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